

Homological Algebra Week2 Solutions

Exercise 4

(1) is a category since G is closed under the operation which admits an identity and is associative.

(2) is a category since a partial ordering is reflexive and transitive. This shows that the identity and the compositions are well-defined. Furthermore, using the transitivity twice, we obtain,

$$(*_{a \leq b} \circ *b \leq c) * \circ *_{c \leq d} = *_{a \leq d} = *_{a \leq b} \circ (*b \leq c * \circ *_{c \leq d}),$$

thus it is associative.